PULSE CODE MODULATION

> The Question will give a range for the sampled amplitudes:

Let's say $-15V$ to $+20V$

> The Question will say how many bits are used:

Let's say 3-bits

Step 1:

Number of zones $= 2^3 = 8$

Step 2:

$$\text{Zone width} = \frac{(+20) - (-15)}{8}$$

$$= 4.375$$

Step 3:

Find the range for the 8 zones.

Step 4:

Find the midpoint for each range.

Step 5:

Assign a Quantization Code to each midpoint / zone

Step 6:

Assign a 3-bit code to each quantization code.

ZONES	MIDPOINTS
+15.625V to +20V	+17.8125V
+11.25V to +15.625V	+13.4375V
+6.875V to +11.25V	+9.0625V
+2.5V to +6.875V	+4.6875V
-1.875V to +2.5V	+0.3125V
-6.25V to -1.875V	-4.0625V
-10.625V to -6.25V	-8.4375V
-15V to -10.625V	-12.8125

Midpoints	Quantization Code	Encoded Words
+17.8125 V	7	111
+13.4375 V	6	110
+9.0625 V	5	101
+4.6875 V	4	100
+0.3125 V	3	011
−4.0625 V	2	010
−8.4375 V	1	001
−12.8125 V	0	000

Now, you are ready to perform PCM on the actual samples.

If a sampled amplitude is let's say +10V.

* Normalized PAM value $= \frac{+10}{\Delta} = \frac{+10}{4.375} = 2.2857\ldots$

* Normalized Quantization value $= \frac{+9.0625}{\Delta} = \frac{+9.0625}{4.375}$

$= 2.0714$

We chose +9.0625 because +10 is in the range +6.875V to +11.25V whose midpoint is +9.0625V

* Error = 2.0714 − 2.2857

= −0.2143

Error is negative. This means the midpoint is below the actual sample.

* Quantization code for +9.0625 is 5

* Encoded word for 5 is 101

In PCM,

- Analog Signal goes into the Sampler.
- PAM Signal comes out.

- PAM Signal is then sent to Quantizer.
- Quantization / Quantized Code comes out.

- Quantized Code is fed to the Encoder.

- Encoded words come out.

This is the Digital Data.
We then use Line Coding to get Digital Signal.